Additional Exercises 5

Improper Integrals & Topology of $\mathbb{R}^n$

Mirrors: List 1 (improper integrals) & Lectures 1 & 3 (topology) | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Determine convergence or divergence

Identify the type (I or II). Find the asymptotic behaviour near the singular point. Compare with the reference p-integral.

(a) ★ $\displaystyle\int_1^\infty \frac{dx}{x^3+1}$

(b) ★ $\displaystyle\int_0^1 \frac{dx}{\sqrt[3]{x}}$

(c) ★ $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,dx$

(d) ★ $\displaystyle\int_0^1 \frac{\sin x}{\sqrt{x}}\,dx$

(e) ★ $\displaystyle\int_0^\infty e^{-x}\,dx$

(f) ★ $\displaystyle\int_2^\infty \frac{dx}{x(\ln x)^2}$

(g) ★★ $\displaystyle\int_1^\infty \left(\sqrt{x+1}-\sqrt{x-1}\right)dx$

(h) ★★ $\displaystyle\int_0^1 \frac{\ln(1+\sqrt[3]{x})}{e^{\sin x}-1}\,dx$

(i) ★★ $\displaystyle\int_0^\infty \frac{\sin x}{x^{3/2}}\,dx$ *(treat $x\to 0^+$ and $x\to\infty$ separately)*

(j) ★★ $\displaystyle\int_0^1 \frac{e^x-1}{\sin^2 x}\,dx$

(k) ★★ $\displaystyle\int_1^\infty \frac{\arctan x}{x^2}\,dx$

(l) ★★★ $\displaystyle\int_1^\infty \left(\sqrt{x^2+1}-\sqrt{x^2-1}\right)dx$

(m) ★★★ $\displaystyle\int_0^1 \frac{\sqrt[3]{x}\ln x}{1-x^2}\,dx$

Exercise 2: Topology of $\mathbb{R}^n$ — classify each set

For each set, identify: (a) whether a given sample point is interior, boundary, or exterior; (b) whether the set is open, closed, both, or neither; (c) whether the set is bounded; (d) whether the set is compact.

(a) ★ $E = \{(x, y) : x^2 + y^2 < 1\}$

(b) ★ $E = \{(x, y) : x^2 + y^2 \leq 1\}$

(c) ★ $E = \{(x, y) : x^2 + y^2 = 1\}$ (the circle itself)

(d) ★ $E = \{(x, y, z) : x^2 + y^2 + z^2 \leq 4\}$

(e) ★★ $E = \{(x, y) : 1 < x^2 + y^2 \leq 4\}$

(f) ★★ $E = \{(x, y) : y > x^2\}$

(g) ★★ $E = \{(x, y) : x > 0,\ y > 0,\ xy \leq 1\}$

(h) ★★ $E = \{(x, y) : |x| + |y| \leq 1\}$

(i) ★★★ $E = \mathbb{Q}^2 = \{(x, y) : x \in \mathbb{Q},\ y \in \mathbb{Q}\}$

(j) ★★★ $E = \{(x, y) : y = \sin(1/x),\ x > 0\}$

Exercise 3: Norms and inequalities

(a) ★ Compute $\|\bar{x}\|_1$, $\|\bar{x}\|_2$, $\|\bar{x}\|_\infty$ for $\bar{x} = (3, -4, 0)$.

(b) ★ Compute $\|\bar{x}\|_1$, $\|\bar{x}\|_2$, $\|\bar{x}\|_\infty$ for $\bar{x} = (1, 2, -2, 0)$.

(c) ★★ Show that for any $\bar{x} \in \mathbb{R}^2$: $\|\bar{x}\|_\infty \leq \|\bar{x}\|_2 \leq \|\bar{x}\|_1$.

(d) ★★ Verify the Cauchy–Schwarz inequality for $\bar{x} = (1, 2, 3)$, $\bar{y} = (4, -1, 2)$.

(e) ★★ Describe the set of points $\bar{x} \in \mathbb{R}^2$ satisfying $\|\bar{x}\|_1 \leq 1$. Sketch it. How does it compare to the unit disk $\|\bar{x}\|_2 \leq 1$?

(f) ★★★ Using the inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$, prove the Cauchy–Schwarz inequality:

$$\left(\int_0^1 f(x)g(x)\,dx\right)^2 \leq \int_0^1 f(x)^2\,dx \cdot \int_0^1 g(x)^2\,dx.$$

Exercise 4: Compute the following improper integrals exactly

Where convergent, find the exact value.

(a) ★ $\displaystyle\int_1^\infty \frac{dx}{x^2}$

(b) ★ $\displaystyle\int_0^1 \frac{dx}{\sqrt{x}}$

(c) ★★ $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\,dx$ *(integrate by parts)*

(d) ★★ $\displaystyle\int_0^\infty x\,e^{-x}\,dx$

(e) ★★★ $\displaystyle\int_0^\infty \frac{dx}{(1+x^2)^2}$ *(use the substitution $x = \tan\theta$)*

★ *p-test or direct substitution* ★★ *Asymptotic equivalence or limit comparison* ★★★ *Rationalization, careful path analysis, or abstract argument.*